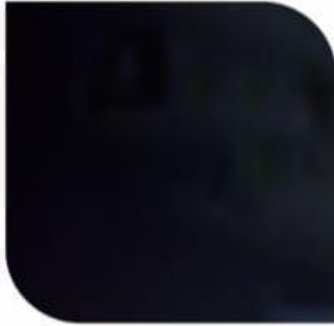


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PROJECT REPORT

TITLE : STOP WATCH

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Submitted to : Global Quest Technologies

Date : 13 June 2026

ABSTRACT

The **Stopwatch** is a Python-based application designed to measure and display elapsed time accurately for various activities such as sports, experiments, and daily tasks. The system provides essential functionalities including start, stop, and reset operations, allowing users to track time intervals efficiently. It continuously updates the elapsed time in real time, offering a simple and interactive user experience.

This project is developed using core Python concepts such as loops, time functions, and conditional statements. The stopwatch records the time when started and calculates the difference when stopped, ensuring precise time measurement. The application can be implemented as a command-line interface (CLI) or enhanced with a graphical user interface (GUI) for better usability.

Overall, the Stopwatch project demonstrates the practical implementation of Python in creating time-based utility applications. It helps in understanding real-time processing and can be further extended with features like lap timing, data storage, and performance tracking, making it a useful tool for both educational and practical purposes.

SOURCE CODE:

```
import sys
from PyQt5.QtWidgets import (QApplication, QWidget, QLabel,
                             QPushButton, QVBoxLayout, QHBoxLayout)
from PyQt5.QtCore import QTimer, QTime, Qt

class Stopwatch(QWidget):
    def __init__(self):
        super().__init__()
        self.time = QTime(0, 0, 0, 0)
        self.time_label = QLabel("00:00:00.00", self)
        self.start_button = QPushButton("Start", self)
        self.stop_button = QPushButton("Stop", self)
        self.reset_button = QPushButton("Reset", self)
        self.timer = QTimer(self)
        self.initUI()

    def initUI(self):
        self.setWindowTitle("Stopwatch")

        vbox = QVBoxLayout()
        vbox.addWidget(self.time_label)

        self.setLayout(vbox)

        self.time_label.setAlignment(Qt.AlignCenter)

        hbox = QHBoxLayout()

        hbox.addWidget(self.start_button)
        hbox.addWidget(self.stop_button)
        hbox.addWidget(self.reset_button)

        vbox.addLayout(hbox)

        self.setStyleSheet("""
            QPushButton, QLabel{
                padding: 20px;

                padding: 20px;
                font-weight: bold;
                font-family: calibri;
            }
            QPushButton{
                font-size: 50px;
            }
            QLabel{
                font-size: 120px;
                background-color: hsl(200, 100%, 85%);
                border-radius: 20px;
            }
        """)

        self.start_button.clicked.connect(self.start)
        self.stop_button.clicked.connect(self.stop)
        self.reset_button.clicked.connect(self.reset)
        self.timer.timeout.connect(self.update_display)

    def start(self):
        self.timer.start(10)

    def stop(self):
        self.timer.stop()

    def reset(self):
        self.timer.stop()
        self.time = QTime(0, 0, 0, 0)
        self.time_label.setText(self.format_time(self.time))

    def format_time(self, time):
        hours = time.hour()
        minutes = time.minute()
        seconds = time.second()
        milliseconds = time.msec() // 10
```

No connection

```

    """
    self.start_button.clicked.connect(self.start)
    self.stop_button.clicked.connect(self.stop)
    self.reset_button.clicked.connect(self.reset)
    self.timer.timeout.connect(self.update_display)

def start(self):
    self.timer.start(10)

def stop(self):
    self.timer.stop()

def reset(self):
    self.timer.stop()
    self.time = QTime(0, 0, 0, 0)
    self.time_label.setText(self.format_time(self.time))

def format_time(self, time):
    hours = time.hour()
    minutes = time.minute()
    seconds = time.second()
    milliseconds = time.msec() // 10
    return f"{hours:02}:{minutes:02}:{seconds:02}.{milliseconds:02}"

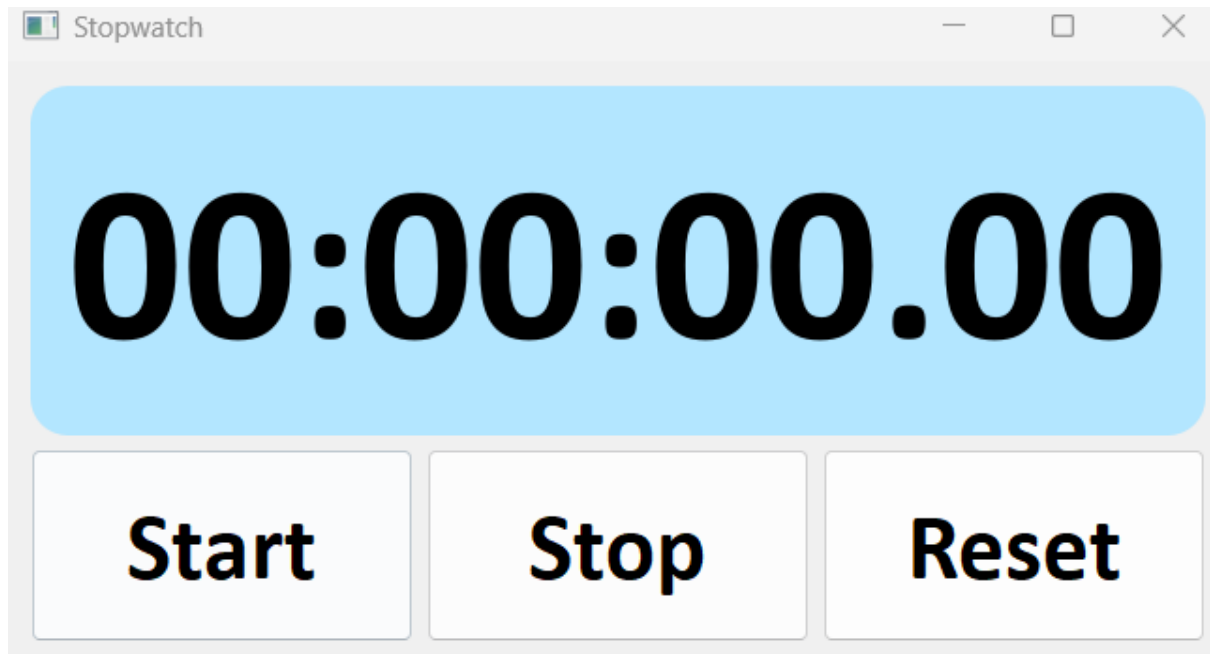
def update_display(self):
    self.time = self.time.addMSecs(10)
    self.time_label.setText(self.format_time(self.time))

if __name__ == "__main__":
    app = QApplication(sys.argv)
    stopwatch = Stopwatch()
    stopwatch.show()
    sys.exit(app.exec_())

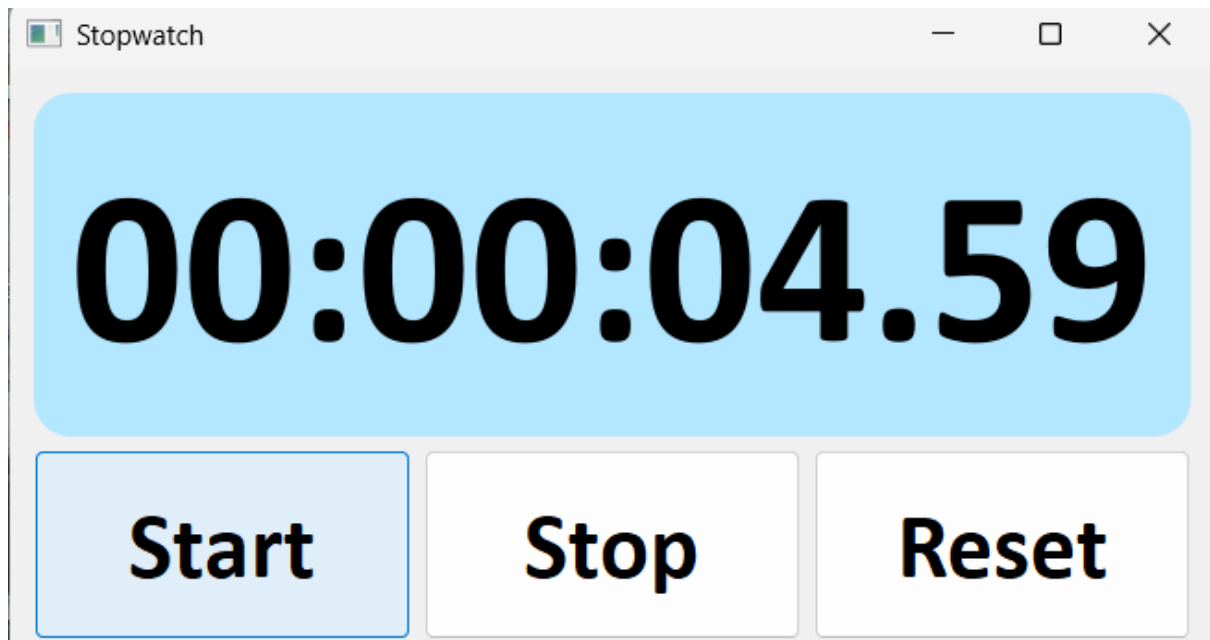
```

OUTPUT:-

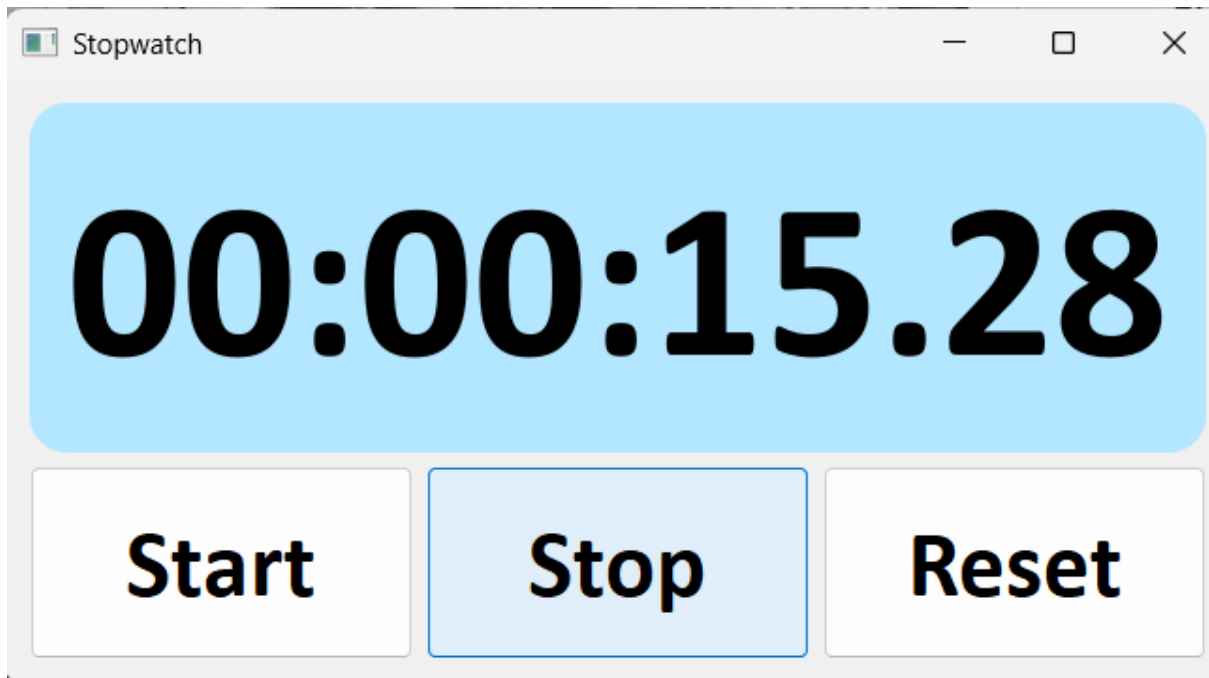
1.Main Menu OutPut



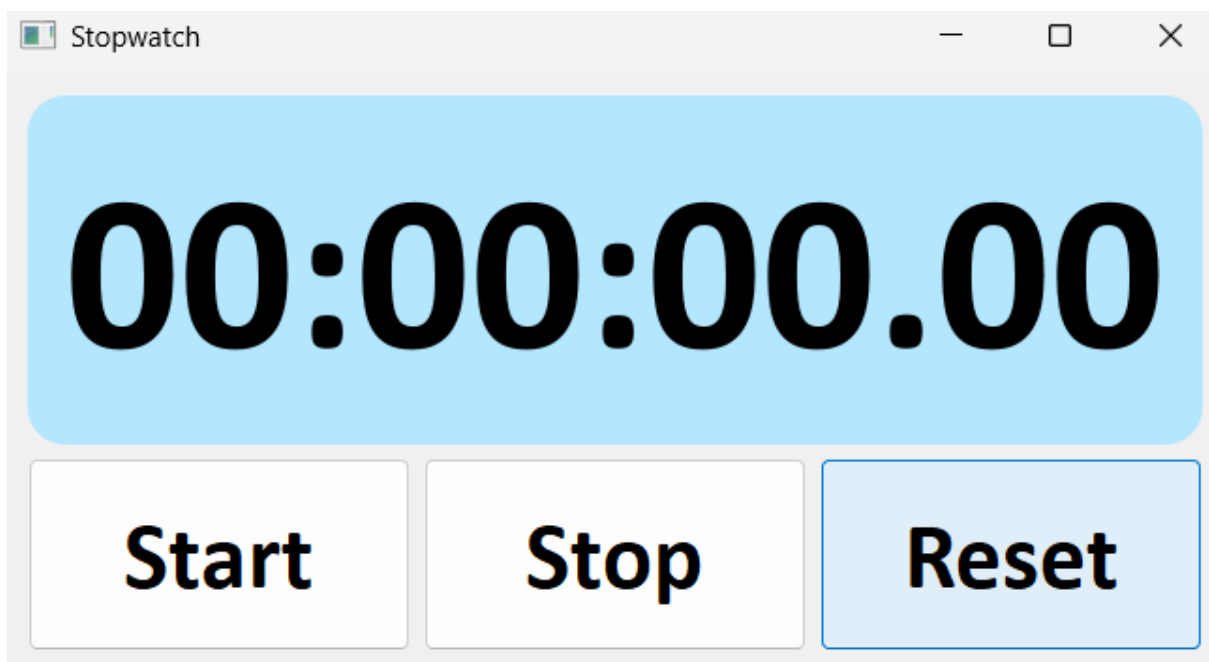
2. Start Output



3. Stop Output



4. Reset Output





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